meters used for measuring length are portions of matter that measure another aspect of its extension.

Spacetime is the gravitational field—and vice versa. It is something that exists by itself, as Newton intuited, even without matter. But it is not an entity that is different from the other things in the world—as Newton believed—it is a field like the others. More than a drawing on a canvas, the world is like a superimposition of canvases, of strata, where the gravitational field is only one among others. Just like the others, it is neither absolute nor uniform, nor is it fixed: it flexes, stretches, and jostles with the others, pushing and pulling against them. Equations describe the reciprocal influences that all the fields have on each other, and spacetime is one of these fields.*

The gravitational field can also be smooth and flat, like a straight surface, and this is the version that Newton described. If we measure it in meters, we discover that the Euclidian geometry that we learned at school applies. But the field can also undulate, in what we call "gravitational waves." It can contract and expand.

Remember the clocks in chapter 1 that slow down in the vicinity of a mass? They slow down because there is, in a precise sense, "less" gravitational field there. There is less time there.

The canvas formed by the gravitational field is like a vast elastic sheet that can be pulled and stretched. Its stretching and bending is the origin of the force of gravity, of things falling, and provides a better explanation of this than the old Newtonian theory of gravity.

Look again at the figure here which illustrates how more time passes above than below, but imagine now that the piece of paper on which the diagram is drawn is elastic; imagine stretching it so that the time in the mountains actually becomes elongated. You will obtain something like the image below, which represents space (the height, on the vertical axis) and time (on the horizontal)—but, now, the "longer" time in the mountains effectively corresponds to a greater length of time.